

CogniCare NER (স্মৃতি সেতু - চক্ষু আঁ)

Offline-First, Culturally-Grounded Multilingual AI Digital Therapeutic
& Clinical Surveillance Ecosystem for Geriatric Cognitive Care

Designed for: Ministry of Development of North Eastern Region (MDoNER) & LGBRIMH Tezpur

EXECUTIVE SUMMARY & SYSTEM PURPOSE

CogniCare NER (স্মৃতি সেতু - চক্ষু আঁ / Smriti-Setu) is an indigenous, clinically-calibrated digital health solution designed to address the critical dementia diagnosis, monitoring, and cognitive rehabilitation gap across the 8 North-Eastern states of India. Operating entirely offline in remote rural hamlets, CogniCare empowers elderly patients, illiterate family caregivers, and grassroots ASHA workers through speech-first local dialects, culturally resonant neuro-therapeutic exercises, computer vision pill tracking, wandering defense, and zero-internet peer-to-peer data synchronization.

Key Platform Attributes & Breakthrough Architecture:

- Target Demographics:** Elderly citizens suffering from Mild Cognitive Impairment (MCI) and Alzheimer's Disease across rural Assam, Meghalaya, Manipur, Mizoram, Nagaland, Tripura, Arunachal Pradesh, and Sikkim.
- Language Inclusivity:** Full localization in 10 languages (Assamese, Bengali, Manipuri, Bodo, Khasi, Mizo, Kokborok, Nepali, Hindi, English) with a regional dialect parser.
- Clinical Foundations:** Automated geometric scoring on the Rouleau Clock Drawing Test (CDT), Sunderland criteria, Hindi Mental State Examination (HMSE-10), MoCA, and Zarit Burden Interview (ZBI-12).
- Zero-Connectivity Core:** PWA local-first architecture with progressive caching, IndexedDB/LocalStorage, and zlib-deflated dynamic optical QR code batch sync.
- Patient Safety Shield:** 500-meter virtual geofencing with audio homing beacon, sundowning circadian lux metering (2700K phototherapy), and blister-pack computer vision verification.
- Caregiver Sustainability:** Acoustic voice check-ins evaluating vocal jitter/shimmer affect flattening paired with a 15-minute guided bamboo flute pranayama decompression player.

Parameter	Specification Details
Hackathon Problem Statement	Smart India Hackathon 2026 — AI Geriatric Care in Tribal & Remote Belts
Lead Partner Institution	Lokopriya Gopinath Bordoloi Regional Institute of Mental Health (LGBRIMH)
Supported States	Assam, Arunachal Pradesh, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim, Tripura
Core Tech Stack	React 19, TypeScript, Vite, Tailwind CSS, Web Audio API, Web Speech API, PWA, pako
Compliance Standards	AB-HWC, Ayushman Bharat Health Account (ABHA), DISHA, HIPAA, WCAG 2.1 AAA
Deployment Model	Zero-Internet Standalone Offline PWA + Asynchronous Firebase Firestore Cloud Mirror

1. The North-East India Geriatric Cognitive Crisis

Understanding the severe healthcare barriers in the 8 North-Eastern states

Dementia and Mild Cognitive Impairment (MCI) represent an escalating yet severely neglected silent epidemic in North-East India (NER). While India has over 8.8 million elderly individuals living with dementia, the North-Eastern region faces profound structural disadvantages that render conventional metropolitan healthcare models completely ineffective.

The Quadruple Barrier to Cognitive Care in NER:

- **Extreme Geoclimatic Remoteness:** Vast tribal populations reside in isolated riverine islands (chars) of the Brahmaputra, steep hills of Nagaland and Mizoram, and dense rain belts of Meghalaya, requiring multiple days of travel to reach district hospitals.
- **Linguistic Diversity & Dialect Drift:** Over 220 indigenous languages and dialects are spoken. Mainstream diagnostic tools in English or standard Hindi completely fail due to phonetic and lexical comprehension barriers.
- **Acute Shortage of Neurologists & Psychiatrists:** Tertiary neuropsychiatric care is almost exclusively centralized at LGBRIMH in Tezpur, Assam, leaving millions without access to early screening or cognitive therapy.
- **Severe Stigma & Cultural Misattribution:** Memory loss and behavioral symptoms of dementia (BPSD) like sundowning or wandering are frequently misattributed to senility ("bura-bhal"), black magic, or spiritual curses, delaying intervention until severe stages.

THE COGNICARE MISSION

CogniCare NER shifts dementia management from reactive hospitalization to proactive, community-based, culturally-affirming micro-interventions. By transforming the everyday smartphone into an offline clinical neuropsychologist and safety shield, CogniCare preserves cognitive dignity at home.

2. End-to-End System Architecture & Local-First Philosophy

Engineered for 100% functionality in zero-bandwidth environments

CogniCare NER is constructed from the ground up on a Local-First, Zero-Internet architecture. The application does not require an active 4G/5G connection to execute neuropsychological assessments, computer vision blister checks, voice synthesizers, or geofence tracking.

Architecture Layer	Technical Implementation	Offline / Resilient Capability
Client Framework	React 19 + TypeScript + Vite	Instant 60fps UI, sub-second boot, tree-shaken bundles
State Management	Zustand with LocalStorage persistence	Zero data loss during unexpected battery shutdowns
Service Worker / PWA	Vite PWA plugin with Workbox caching	Complete caching of scripts, fonts, audios, and icons
Audio Synthesizer	Web Speech API + Web Audio API	Offline voice prompting & procedural binaural frequencies
Computer Vision	HTML5 Canvas 2D image pixel processing	In-browser blister pack cavity and contour detection
P2P Optical Sync	Zlib Deflate (pako) + Animated QR Stream	Batch transmission of 50+ records across devices in 15s
Cloud Telemetry	Firebase Cloud Firestore (Optional / Async)	Bidirectional cloud backup whenever cellular towers link

3. Multilingual Engine & Regional Dialect Code-Mixing

Breaking tribal language barriers across 10 North-Eastern tongues

Cognitive evaluation is deeply entwined with language. Standard cognitive exams like the Mini-Mental State Examination (MMSE) produce artificially depressed scores when administered to rural elders in non-native languages. CogniCare implements a bidirectional localization engine across 10 official languages and a real-time dialect parser.

- **Assamese (... , Deôrákô)** : Default regional tongue for the Brahmaputra Valley, with phonetically tuned speech synthesis.
- **Bengali & Sylheti (... , Sylheti)** : Targeted for the Barak Valley (Cachar, Karimganj, Hailakandi) and Tripura plains.

- **Manipuri / Meitei** (,Í®Ã¤¿,-Ç²Á): Localized for the Imphal Valley, accommodating Meitei Mayek script and terminology.
- **Bodo** (,Í®Ã¤¿,-Ç²Á): Localized for the Bodoland Territorial Region (BTR), using Devanagari script adaptations.
- **Tribal Languages**: Full semantic translation for Khasi (Meghalaya), Mizo (Mizoram), Kokborok (Tripura), and Nepali (Sikkim/Darjeeling).

Dialect Code-Mixing Natural Language Parser (dialectParser.ts):

Rural elders rarely speak standard textbook Assamese or Bengali; instead, they code-mix regional dialects (e.g., Kamrupi in Lower Assam, Goalpariya in Western Assam, Sylheti in Southern Assam, or Bodo-Assamese in northern plains). CogniCare integrates a specialized regional synset dictionary and Levenshtein phonetic distance matcher ($D \leq 2$).

When an elder speaks "–¾đÁ–¾" (kharuba) instead of the standard "–¾đ" (khar - traditional curry) during the Food Association exercise, the parser detects the cultural term, resolves it to its canonical clinical category, displays a regional dialect recognition badge, and awards full semantic fluency points without penalizing the patient.

4. Complete Neuro-Therapeutic Suite: 8 Culturally Grounded Games
Rigorous cognitive domain stimulation through familiar North-East cultural touchstones

Cognitive stimulation therapy (CST) is clinically proven to slow cognitive decline in mild-to-moderate dementia. However, games built on western archetypes (e.g. chess, poker, western foods) cause severe alienation and confusion. CogniCare re-engineers CST using 8 deeply familiar North-Eastern cultural traditions:

1. Gamosa Card Match (,Í®Ã¤¿ -Ç² — Visual Working Memory)

The patient matches pairs of traditional handwoven Assamese Gamosa red-and-white border motifs, including Phulam Gamosa (floral weave), Kingkhap (royal Ahom peacock motif), Mising tribal geometric weaves, Karbi motifs, Bodo weaves, and the Bihu Dhol drum. The exercise scales from 4 to 8 pairs with soothing Assamese voice encouragement.

2. Bihu Rhythm Recall (–¿¹Á >´Í! ,Í®Ã¤¿ — Auditory Working Memory)

Stimulates the temporal auditory cortex using traditional North-East percussion instruments: Dhol (two-headed drum), Pepa (buffalo-horn trumpet), Taal (bronze cymbals), Gogona (bamboo jaw harp), Toka (bamboo clapper), and Khol. The system plays sequential rhythmic patterns that the elder replicates, exercising temporal sequence processing and auditory working memory.

3. Daily Routine Sequencer (¡È´´Í!¿´ •¾đÍ–Í•Íđ® — Executive Function)

Executive dysfunction causes dementia patients to lose the ability to sequence daily tasks. This module asks the patient to order cards depicting a typical village daily routine: waking up and drinking morning red tea (Lal Cha), bathing at the pond/tubewell, praying at the village Namghar or temple, tending the home garden, taking afternoon rest, and taking evening medication.

4. Bamboo Pattern Completion (–¾•¹đ †đÍ¿ -Ç² — Visuospatial Reasoning)

Draws inspiration from the intricate bamboo and cane weaving matrices (Japi hats, fishing traps like Polo and Jakoi) crafted by North-East artisans. The elder analyzes a missing slot in a geometric matrix and selects the matching pattern, exercising parietal lobe visuospatial reasoning and inductive logic.

5. Market Spot Difference (¹¾œ¾đ –œ¾đ — Selective Visual Attention)

Presents dual illustrations of a vibrant rural North-East weekly market (haat) featuring vegetable stalls with local greens (Dhekia, Lai Xak), pottery vendors, and fish sellers. The patient must locate subtle differences, training selective visual attention, visual search speed, and inhibitory control.

6. Word Association & Food Categorization (–¾!Í– ¶–Í! ,–È– — Semantic Memory)

Prompts the elder to name traditional North-East delicacies: Khar (alkaline preparation), Tenga (sour fish curry), Pitha (rice cakes), Chira-Doi (flattened rice with curd), Laru, and Jolpan. Integrated with speech-to-text and the Dialect Parser, it tests lexical retrieval and language semantic networks.

7. Digital Clock Drawing Test with 3-Layer AI Scoring (~¡¾À ...•¾ ºđÀ•)

The Clock Drawing Test (CDT) is a gold-standard global clinical instrument for dementia screening, evaluating visuospatial, executive, and abstract conceptualization domains. CogniCare implements a digital touch canvas where the elder is instructed to draw a clock face, place numbers 1 to 12, and set the hands to "10 past 11" (11:10).

Our proprietary 3-Layer Geometric AI evaluates the drawing in real time:

- **Layer A (Contour Circularity):** Analyzes bounding-box aspect ratio (H/W), contour closure gap (< 35px), and perimeter smoothness to detect visuospatial distortion.
- **Layer B (Number Placement Symmetry):** Computes centroid coordinates across all 4 quadrants (Q1-Q4). Crucially, detects Hemispatial Neglect (where all numbers are scrunched onto the right half of the clock face, signaling contralateral parietal lobe neurodegeneration).
- **Layer C (Hand Angle Accuracy):** Vector trigonometry calculates angles for the hour hand (target: 300 deg +/- 15 deg) and minute hand (target: 60 deg +/- 15 deg) relative to the pivot center, while penalizing perseveration (extra hands).

Outputs standard Rouleau Scale (1 to 5) and Sunderland Composite Score (1 to 10) for direct clinical review.

8. Olfactory (Aroma) Memory Recall Kit (,Á—“Í\$¿,Í@Ã¤¿ªðÀ•Í·¾)

The olfactory bulb and entorhinal cortex are among the very first brain structures damaged by Alzheimer's neurofibrillary tangles, often up to 5 to 8 years before clinical memory loss manifests. CogniCare's Olfactory Recall Kit leverages 5 culturally distinctive physical scent vials provided in an ASHA field kit:

- **Kazi Nemu:** Indigenous Assam oval lemon with pungent citrus volatile terpene profile.
- **Chai Paat:** Fresh, bruised green tea garden leaves with earthy herbal notes.
- **Bhut Jolokia:** Smoked King Chilli aroma activating trigeminal sensory pathways.
- **Mati Gondho:** Petrichor — scent of dry soil drenched by the first monsoon rains.
- **Joha Chaul:** Assam indigenous aromatic scented rice containing 2-acetyl-1-pyrroline.

The application guides the elder via voice prompts to sniff the numbered vial and select the corresponding memory or picture, computing a Cross-Modality Human Recall Index (HRI) for ultra-early pre-symptomatic detection.

5. Patient Safety, BPSD Management & Assistive Care Modules

Protecting vulnerable elders from wandering, sundowning agitation, and medication hazards

1. Safe Home SOS & Virtual Geofence Wandering Defense (,Áð•Í·¿¤~ð)

Wandering is one of the most hazardous behavioral manifestations of dementia, leading to severe injury or death in rural riverine and dense jungle terrains. CogniCare implements an autonomous geofencing defense system centered on the elder's home coordinates (calibrated to Bihaguri Gaon, Sonitpur):

- **500-Meter Haversine Geofence:** Continuously calculates great-circle distance between live GPS coordinates and home safezone using spherical trigonometry ($R = 6,371$ km).
- **High-Contrast Safe Return Mode:** If the elder is confused outdoors, opening the app presents an ultra-large directional compass pointing directly toward home with a distance countdown in Assamese/English.
- **Acoustic Homing Beacon:** Synthesizes an alternating dual-frequency audio tone (880Hz / 440Hz) via the Web Audio API, helping nearby villagers locate the lost elder.
- **Automatic WhatsApp Alert Dispatch:** A geofence breach immediately constructs an encrypted WhatsApp alert with exact latitude/longitude coordinates and a clickable Google Maps navigation link sent to the primary caregiver and village ASHA worker.

2. Sundowning Calm & Circadian Ambient Lux Meter (—\$Â²¿ðªÍð¶¾“Í¤¿

Sundowning — extreme anxiety, pacing, confusion, and aggression occurring as daylight wanes between 4:30 PM and 7:00 PM — severely destabilizes dementia households. CogniCare addresses sundowning through ambient physics and phototherapy:

- **Photometric Camera Lux Meter:** Samples ambient luminance from the device camera video feed using the ITU-R BT.709 relative luminance formula: $Y = 0.2126R + 0.7152G + 0.0722B$. Calibrated to identify Twilight Dim Light (< 150 lux) as a critical sundowning trigger.
- **2700K Warm Amber Phototherapy:** Transforms the smartphone/tablet screen into a warm 2700K amber glow panel to reset suprachiasmatic circadian pacemaker rhythms and inhibit cortisol spikes.
- **Therapeutic Ambient Soundscapes:** Plays procedural alpha-wave binaural pulses (432Hz) layered with authentic North-Eastern acoustic textures: gentle Brahmaputra river currents, monsoon rain on tea bushes, and evening bamboo flutes.

3. Camera Blister-Pack Medication Verification (”·\$ªðÀ•Í·¾)

Medication non-adherence and accidental double-dosing of cholinesterase inhibitors (e.g. Donepezil, Memantine) or antihypertensives pose lethal risks. CogniCare's Blister-Pack Computer Vision scanner allows the elder or caregiver to hold their foil blister pack up to the camera. Using canvas pixel variance and luminance standard deviation, the system differentiates intact shiny foil bubbles from punctured, dark cavity depressions, automatically verifying that today's dose was safely taken and issuing a stark red warning if an overdose is detected.

4. Reminiscence Digital Album (,Í®Ã¤¿ •²¬¾®)

Curates family photographs, ancestral village scenes, and youth milestones coupled with recorded voice notes from grandchildren and siblings. By activating spared remote retrograde episodic memory, it grounds agitated elders during moments of profound dissociation.

6. Caregiver Intelligence, Burnout Respite & Trajectory AI
Protecting family caregivers from severe emotional exhaustion while forecasting progression

1. Predictive AI Cognitive Trajectory Forecaster (12-Month Projections)

Dementia care is plagued by uncertainty. Family members often do not know whether interventions are working or what the next year holds. CogniCare introduces a dual-curve trajectory forecaster using historical session analytics, cognitive domain variance, and baseline decline slopes:

- **Curve A: Natural Unintervened Decline:** Depicts the projected sharp drop in MoCA and composite cognitive scores without cognitive stimulation or lifestyle intervention.
- **Curve B: Multimodal Intervened Pathway:** Forecasts the preserved cognitive plateau achievable through daily CST workouts, physical activity, and dietary neuroprotection.
- **North-East Lifestyle AI Prescriptions:** Recommends evidence-based local nutritional and motor-auditory interventions: Chak-hao (Manipuri Black Rice) anthocyanin neuroprotection, morning Assam CTC polyphenol antioxidant regimens, and 15-minute Bihu dhol rhythm motor-auditory synchronization.

2. Caregiver Acoustic Burnout & Respite AI (ZBI-12)

Caregiver burden is a leading driver of premature elder abandonment and institutionalization. CogniCare protects caregivers through acoustic voice check-ins:

- **45-Second Voice Check-in:** Caregiver speaks into the microphone describing their day. The system processes vocal biomarkers: micro-pitch jitter (F0 instability), acoustic shimmer (amplitude perturbation), pitch standard deviation (flattened affect), and sigh/pause density.
- **Zarit Burden Score (ZBI-12):** Calculates clinically calibrated Zarit score (0 to 24), stratifying burnout into Low, Moderate, or Severe.
- **15-Minute Flute & Pranayama Player:** Built-in mindfulness player guiding 4-4-6 rhythmic breathing (4s inhale, 4s hold, 6s exhale) with soothing acoustic North-East bamboo flute harmonics.
- **Emergency Family Respite Alert:** If severe burnout is detected, automatically drafts a WhatsApp message to secondary relatives requesting immediate 2-hour respite relief.

7. ASHA Worker Rural Gateway & Zero-Internet Optical Sync
Equipping grassroots health workers with rapid screening and offline data sharing

Accredited Social Health Activists (ASHAs) are the operational backbone of India's rural public health apparatus. CogniCare's ASHA Screening Portal equips these frontline workers with digital screening batteries and a world-first optical synchronization mechanism:

Standardized Field Screening Batteries:

- **Hindi Mental State Examination (HMSE-10):** Culturally adapted 10-point screening battery calibrated for rural and low-literacy elders.
- **Geriatric Depression Scale (GDS-4):** Rapid 4-question depression screen distinguishing dementia-related apathy from clinical depression (pseudodementia).
- **Activities of Daily Living (ADL / IADL):** Evaluates eating, bathing, dressing, medication management, and village market navigation independence.

- **eSanjeevani Teleconsultation Generator:** Automatically compiles screening findings, CDT scores, and patient ABHA IDs into a standardized referral dossier for instant transmission to LGBRIMH geriatric specialists.

Zero-Internet Peer-to-Peer Batch QR Optical Handoff:

When an ASHA worker screens 30 patients across deep-jungle villages without cellular connectivity, she cannot upload records to the cloud. When she meets an ANM or Medical Officer at the Primary Health Centre (PHC), CogniCare uses a high-speed optical transfer protocol:

- **Step 1: Zlib Deflate Compression:** Multiple patient records and CDT metrics are serialized into JSON and compressed with zlib deflate (pako level 9), reducing payload size by ~75%.
- **Step 2: Base64 Chunking:** The binary payload is sliced into sequential packets of ~650 characters, formatted with packet headers: { id, idx, total, chunk }.
- **Step 3: Animated QR Streamer:** The ASHA's phone screen cycles through dynamic QR codes on an HTML5 canvas at 3 to 5 frames per second.
- **Step 4: Optical Camera Scanner:** The receiver's phone camera scans the streaming QR code, tracks missing frames in a bitmask, and automatically reassembles and decompresses the database upon completion.

This allows 50+ complete patient medical files to be transferred across devices in 15 seconds without internet, Bluetooth pairing, Wi-Fi Direct, or cables!

8. Non-Technical Step-by-Step User Manual

How any user, judge, clinician, or family member can navigate CogniCare NER

Role / Step	Target Action	Expected System Behavior
1. Launch App	Open browser / PWA icon	Loads instantly offline. Prompts role selection (Patient / Caregiver / ASHA).
2. Language Select	Tap globe icon in header	Select from 10 North-East languages. Full interface updates instantly.
3. Patient Workout	Tap any game card (1 to 8)	Speech prompt begins in local tongue. Exercises memory, visuospatial, language.
4. Clock Draw Test	Tap Card #7 (Clock Drawing)	Draw circle, numbers, and hands at 11:10. Tap Submit to see 3-Layer AI scores.
5. Pill Check	Tap "Verify Pill (Camera)"	Hold blister pack to screen. Camera counts empty cavities and confirms dose.
6. Wandering SOS	Tap "Safe Home SOS"	Radar displays home proximity. Tap "Simulate Breach" to see compass & WhatsApp dispa
7. Caregiver Voice	Dashboard -> Respite tab	Record 45s voice. Evaluates vocal jitter & suggests 15-min flute pranayama.
8. Trajectory AI	Dashboard -> Trajectory tab	View 12-month dual decline curves and North-East diet prescriptions.
9. ASHA Batch Sync	ASHA Portal -> Batch QR tab	Generate animated QR on device A; point camera on device B to import records offline.

9. Security, Data Privacy & Clinical Validation Standards

Adhering to Indian and global healthcare compliance benchmarks

- **Local-First Biometric Privacy:** Audio recordings for vocal biomarker analysis and camera frames for blister-pack scanning are strictly processed in-memory inside the client browser. No raw biometrics are ever uploaded to cloud servers.
- **Ayushman Bharat & ABHA Alignment:** Standardized patient health identifiers compatible with National Health Authority (NHA) protocols and eSanjeevani digital OPD.
- **Accessibility Compliance (WCAG 2.1 AAA):** Designed with minimum 48px touch targets, high-contrast warm color palettes, and full voice navigation for illiterate rural elders.
- **DISHA & HIPAA Architecture:** Encrypted local storage with SHA-256 integrity verification across P2P optical payloads.

10. Project Summary & National Scaling Roadmap

Transforming geriatric cognitive healthcare across India's frontiers

CogniCare NER (,Í®Ã¤¿-,Ç¤Á) demonstrates how cutting-edge digital health technology can leverage India's rich cultural heritage to solve India's toughest public health challenges. By removing the twin dependencies on internet connectivity and English/Hindi language fluency, CogniCare brings tertiary neuropsychiatric care directly into the bamboo homes and rural tea gardens of North-East India.

FUTURE ROADMAP & MDoNER SCALING

Phase 1 (SIH 2026): Pilot testing in Sonitpur and Kamrup districts, Assam in collaboration with LGBRIMH Tezpur.
Phase 2 (Year 1): Deployment across 1,000 Ayushman Bharat Health & Wellness Centres (AB-HWCs) across all 8 North-Eastern states.
Phase 3 (Year 2): Integration with National Teleconsultation Service (eSanjeevani) and expansion into Central and Himalayan tribal belts.